

APPENDIX A

Math

This appendix provides an introduction to the mathematics used in spacecraft simulation and control-system design. The topic of numerical integration is covered because virtually all simulations of spacecraft require numerical integration. Matrices and vectors are important elements needed to model spacecraft. Spherical geometry is covered along with some elements of calculus.

A.1. Vectors and matrices

A.1.1 Notation

In this book, vectors may represent either ordered collections of values, or a quantity with a magnitude and direction in a 3-space. The state vector for a system is an example of the former; the position vector of a thruster is an example of the latter. The state vector contains all quantities to represent the time-varying quantities of a system. For a point mass these would be position and velocity. For a circuit it would be the currents in the wires.

This book uses a combination matrix and vector notation to represent vectors and matrices. Matrix notation implies that each vector is connected with a specific coordinate frame. Thus the position vector

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad (\text{A.1})$$

cannot be assigned a value unless the reference frame for which x , y , and z are defined is given. Usually the reference frame will be obvious from the context. For example, if the transformation matrix B transforms from frame b to a , then

$$y = Bu \quad (\text{A.2})$$

implies that u is in frame b and y is in frame a . Note the use of a capital letter for the matrix. In some cases, mostly when presenting other authors' work, vector notation will be used and will be clear from the context. A vector, in contrast to a " 3×1 matrix" exists independently of any coordinate frame conceptually, but when numbers are to be used the coordinate frame must be specified.

Subscripts will be used to denote the elements of a vector. For example, the vector v is

$$v = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} \quad (\text{A.3})$$

and the matrix m might be

$$m = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} \quad (\text{A.4})$$

Vectors are single-column matrices. Sometimes, however, subscripts are used to differentiate one vector from another. In those cases, the second element of the subscript will denote the vector component.

$$v_1(x) = \begin{bmatrix} v_{1x}(x) \\ v_{1y}(x) \\ v_{1z}(x) \end{bmatrix} \quad (\text{A.5})$$

In this case the vector is a function of x , the x does not denote element x .

Row vectors are obtained by taking the transpose of a column vector.

$$a = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} \quad (\text{A.6})$$

$$a^T = \begin{bmatrix} a_x & a_y & a_z \end{bmatrix} \quad (\text{A.7})$$

This convention differs from other books where a vector may be a row or column vector depending on the context.

A matrix may be composed of other matrices. For example, a matrix composed of three vectors is

$$\begin{bmatrix} u_x & u_y & u_z \end{bmatrix} = \begin{bmatrix} u_{xx} & u_{yx} & u_{zx} \\ u_{xy} & u_{yy} & u_{zy} \\ u_{xz} & u_{yz} & u_{zz} \end{bmatrix} \quad (\text{A.8})$$

Each column is a vector. The first index refers to the vector and the second to the component of the vector.

A.1.2 Vector and matrix representations of operations

A few common vector operations will be described here, along with their corresponding matrix representation. In this section, vectors will be denoted with an arrow.

The dot product (or inner product) of a vector is equivalent to the transpose of a 3×1 matrix times a 3×1 matrix.

$$\vec{a} \cdot \vec{b} = a^T b = |a||b|\cos\theta \quad (\text{A.9})$$

where θ is the angle between the two vectors. The crossproduct of two vectors is equivalent to the product of the skew-symmetric form of the column matrix times the second vector.

$$\vec{a} \times \vec{b} = a^\times b \quad (\text{A.10})$$

where the skew-symmetric form of a is

$$a^\times = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix} \quad (\text{A.11})$$

Note that this particular skew-symmetric matrix is not invertible. This matrix will come into play in several areas in this book.

The last vector operation is also known as the outer product.

$$\vec{a}\vec{b} = ab^T = \begin{bmatrix} a_x b_x & a_x b_y & a_x b_z \\ a_y b_x & a_y b_y & a_y b_z \\ a_z b_x & a_z b_y & a_z b_z \end{bmatrix} \quad (\text{A.12})$$

Throughout this book the identity matrix will be denoted by the symbol E , instead of I so that it is not confused with the rotational inertia matrix. The arrow notation denotes the quantity as a vector that is not tied to a particular coordinate frame.

A.1.3 Matrix operations

Matrix addition and subtraction are defined for matrices of the same size

$$A = B + C \quad (\text{A.13})$$

Matrix multiplication is defined for matrices as follows

$$A(n, p) = B(n, m)C(m, p) \quad (\text{A.14})$$

Matrix multiplication is associative

$$A(BC) = (AB)C \quad (\text{A.15})$$

and distributive

$$A(B + C) = AB + AC \quad (\text{A.16})$$

Generally, however

$$AB \neq BA \quad (\text{A.17})$$

The transpose of a matrix is

$$A = B^T, A_{ij} = B_{ji} \quad (\text{A.18})$$

and

$$(AB)^T = B^T A^T \quad (\text{A.19})$$

A.1.4 Special matrices

A matrix is symmetric if

$$A = A^T \quad (\text{A.20})$$

A matrix is skew-symmetric if

$$A^T = -A \quad (\text{A.21})$$

The matrix form of the vector crossproduct is a skew-symmetric matrix, but the diagonal does not have to be zero for a matrix to be skew-symmetric.

A matrix is orthonormal if the rows and columns are orthogonal and the magnitude of each row and column is one. This is true for transformation matrices.

A.1.5 Useful matrix–vector identities

The following is a list of useful matrix–vector identities given without proof. In the following equations, C is an orthonormal matrix (each row and column is a unit vector). A and B are real matrices, and a and b are vectors (i.e., column matrices).

$$\begin{aligned} C^{-1} &= C^T \\ -a^\times b &= b^\times a \\ (Ca)^\times &= Ca^\times C^T \\ -a^\times b^\times b^\times a &= b^\times a^\times a^\times b \\ (A+B)^T &= A^T + B^T \\ (ab)^T &= b^T a^T \\ (a^\times)^T &= -a^\times \\ (A^T)^{-1} &= (A^{-1})^T \\ (AB)^{-1} &= B^{-1} A^{-1} \\ |AB| &= |A||B| \\ A(BD) &= (AB)D \end{aligned}$$

$$\begin{aligned}
 |a| &= \sqrt{a^T a} \\
 \text{Unit}(a) &= \frac{a}{|a|} \\
 \cos \theta &= \frac{a^T b}{|a||b|}
 \end{aligned}$$

A.2. Numerical integration

Spacecraft control-system designers rely on simulations to verify their designs prior to the construction and operation of prototypes. The term “Digital Twin” is sometimes used. Once a spacecraft is in operation, simulations are useful for studying anomalous behavior and in improving the designer’s physical understanding of the device. A simulation attempts to replicate the important behavior of a system. It is not possible to represent a system exactly.

When we talk about simulation, it is convenient to break it into two categories, linear and nonlinear. Almost all real physical devices are nonlinear. Nonetheless, it is usually possible to linearize the dynamics and devices about some operating point where, in a sufficiently restricted region, the system behaves linearly. Sometimes, control designers take advantage of nonlinearities, but in many cases designs are for operating points where the system, if deviations from the operating point are small, is linear. For example, a nonlinear function is

$$y = x^3 \tag{A.22}$$

If we take the derivative at $x = x_0$ we get a linearized version

$$y = x_0^3 + 3x_0^2(x - x_0) \tag{A.23}$$

The nonlinear and linear systems coincide at $x = x_0$ as shown in Fig. A.1. The linear approximation is close to the nonlinear between $0.9 < x < 1.1$.

When dealing with the linearized system, we can work with a state-space description of the form:

$$\dot{x} = ax + bu \tag{A.24}$$

$$y = cx + du \tag{A.25}$$

where a is the plant matrix that relates the state x to the state derivative, b is the input matrix that relates the inputs, such as external disturbances and control inputs, to the state derivative, y is the measured output of the system, c relates the state to the measurements; d provides for direct feedthrough of inputs into the measurements, and u is the input. Often d is omitted, but actually it appears frequently in practice. For example,

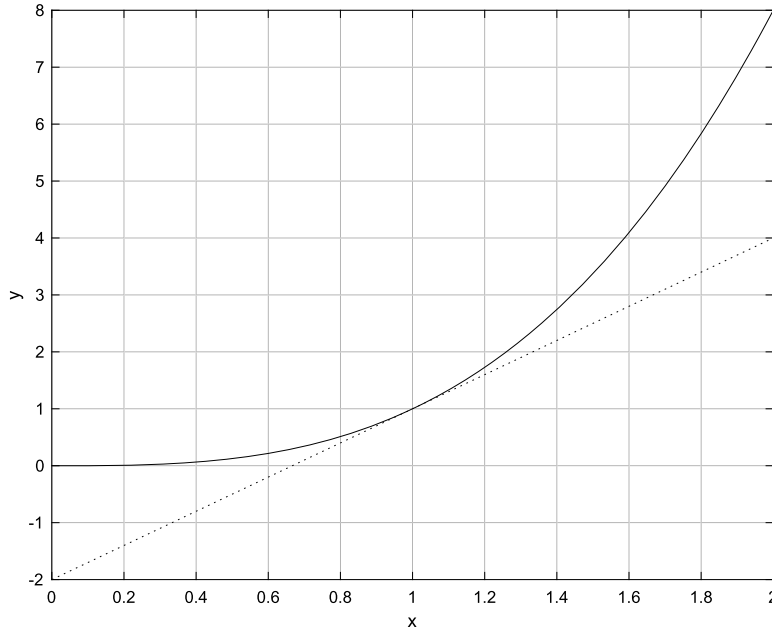


Figure A.1 Linearization of the function $y = x^3$ around $x = 1$.

the state-space description of the system

$$\dot{y} = y + \dot{\phi} \quad (\text{A.26})$$

is

$$\dot{x} = x + \dot{\phi} \quad (\text{A.27})$$

$$y = x + \phi \quad (\text{A.28})$$

where $d = 1$. Sometimes the state is completely measured and c is the diagonal matrix. In that case, the measurement equation is often omitted completely. Some control-design tools cannot handle systems with a nonzero d matrix.

A.2.1 Linearizing a system

Given a system of the form

$$\dot{x} = f(x, u, t) \quad (\text{A.29})$$

$$y = g(x, u, t) \quad (\text{A.30})$$

it can be linearized about (x_0, u_0, t_0) by expanding the function in a Taylor series to first order

$$\dot{x} = \frac{d}{dx}(f(x, u, t))_{(x_0, u_0, t_0)} + \frac{d}{du}(f(x, u, t))_{(x_0, u_0, t_0)} + f(x_0, u_0, t_0) \quad (\text{A.31})$$

$$\gamma = \frac{d}{dx}(g(x, u, t))_{(x_0, u_0, t_0)} + \frac{d}{du}(g(x, u, t))_{(x_0, u_0, t_0)} + g(x_0, u_0, t_0) \quad (\text{A.32})$$

or

$$\dot{x} = ax + bu + f_0 \quad (\text{A.33})$$

$$\delta\gamma = cx + du + g_0 \quad (\text{A.34})$$

If the equations are linearized about an equilibrium state, then f_0 is zero.

The simplest way to simulate a continuous-time system is to discretize it using the zero-order hold, of which there are two types. One is the standard zero-order hold. A second is the delta form of the zero-order hold. These approximations assume that the input is held constant over the interval T . This approximation is satisfactory for most spacecraft control-design problems. An alternative approach is to discretize the system using a first-order hold. This approximation assumes that the input varies linearly from step k to step $k + 1$.

Problems can arise when the dynamical system has discontinuities. For example, friction is often modeled with a Coulomb term that jumps from a negative force or torque to a positive force or torque at zero speed. If the system is linearized about this point, this effect will not be seen. Sometimes this can be solved by using a better model of the system since some discontinuities are really just first-order approximations. In the case of Coulomb friction, one can replace it with a model that has a finite slope at zero or with a bristle-friction model that has an additional friction state.

A.2.2 Nonlinear

Nonlinear systems usually require numerical integration to get a solution. This often means by Runge–Kutta integration. Assume that the system is formulated as

$$\dot{x} = f(x, u, t) \quad (\text{A.35})$$

$$\gamma = g(x, u, t) \quad (\text{A.36})$$

and f is called by the integrator. All continuous models must be included in f . Discrete models, for example the digital control system, are outside of the numerical-integration routine.

Dynamics and control deals primarily with the initial-value problem.

$$\dot{x} = f(x, u, t)$$

$$\begin{aligned} y &= g(x, t) \\ x(t_0) &= x_0 \end{aligned} \tag{A.37}$$

A.2.2.1 Numerical integration methods

Assume that our problem is to solve the linear differential equation

$$\dot{x} = -x + u \tag{A.38}$$

The analytical solution is

$$x(t) = x(0)e^{-t} + (1 - e^{-t})u(t) \tag{A.39}$$

If Δt is small we might try truncating the Taylor series to first order.

$$x(t+h) = x(t) + ht(u(t) - x(t)) \tag{A.40}$$

This is known as Euler integration. By setting $t = 0$ we can compare it against the analytical solution

$$x(h) = x(0)e^{-h} + (1 - e^{-h})u(0) \tag{A.41}$$

$$x(h) = x(0)(1 - h) + hu(0) \tag{A.42}$$

which is correct since the series expansion of e^{-t} to first order is $1 - t$.

An alternative method is to compute the value of x at the time step $h/2$. This is a 2nd-order Runge–Kutta method. We write

$$x(t+h) = x(t) + hf(x(t+h/2), u(t+h/2), t+h/2) \tag{A.43}$$

$$x(t+h/2) = x(t) + \frac{h}{2}f(x(t), u(t), t) \tag{A.44}$$

The solution is

$$x(h) = x(0)\left(1 - h + \frac{h^2}{2}\right) - \frac{h^2}{2}u(0) + hu\left(\frac{h}{2}\right) \tag{A.45}$$

The approximation to the exponential is now the order of h^2 . However, we need to know u at the time step $h/2$, which may be a disadvantage. This is not the only type of 2nd-order Runge–Kutta method but all involve two evaluations of f .

The standard 4th-order Runge–Kutta formula is:

$$\begin{aligned} k_1 &= f(x, u(t), t) \\ k_2 &= f\left(x + \frac{h}{2}k_1, u\left(t + \frac{h}{2}\right), t + \frac{h}{2}\right) \end{aligned}$$

$$\begin{aligned}
 k_3 &= f\left(x + \frac{h}{2}k_2, u\left(t + \frac{h}{2}\right), t + \frac{h}{2}\right) \\
 k_4 &= f\left(x + hk_3, u(t + h - \epsilon), t + h\right) \\
 x &= x + \frac{h}{6}(k_1 + 2(k_2 + k_3) + k_4) + O(h^4)
 \end{aligned}
 \tag{A.46}$$

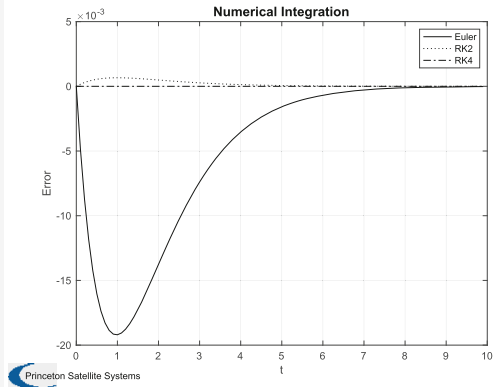
The right-hand sides are computed at four different points, twice at $h/2$ and once at t and $t + h$. At $t + h$ the input term is given at $t + h - \epsilon$, where ϵ is an infinitesimal number. This means that if u is changing at every integration time step (as a digital controller might) then the previously held value is used, not the new value.

The three methods are compared in the following example, where $u = 1$ and $x(0) = 2$. All methods converge to the correct solution but the Euler method has a large initial error.

```

1 n = 100;
2 p = zeros(3,n);
3 x1 = 2; x2 = 2; x3 = 2;
4 h = 0.1;
5 for k = 1:n
6     p(:,k) = [x1;x2;x3];
7     x1 = Euler(@RHSLinear, x1, h, 0);
8     x2 = RK2(@RHSLinear, x2, h, 0);
9     x3 = RK4(@RHSLinear, x3, h, 0);
10 end
11 t = (0:(n-1))*h;
12 e = 2*exp(-t) + 1-exp(-t);
13 Plot2D(t,p - [e;e;e], 't', 'Error', 'Numerical_
    Integration');
14 legend('Euler', 'RK2', 'RK4');
15
16 %% Right hand side
17 function xDot = RHSLinear( x, t )
18
19 xDot = 1 - x;
20
21 end
22
23 function x = Euler(Fun,x,dT,t,varargin)
24
25 x = x + dT*feval(Fun,x,t,varargin{:});
26
27 end

```



Example A.1: Numerical integration.

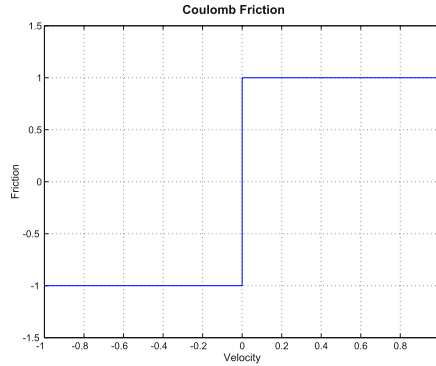
This method does not provide an error estimate. However, it is possible to run a 4th- and 5th-order (or in general n - and $n + 1$ -order) method with only one more calculation of f . The difference between the solutions can be used to adjust the step size. Since the step size is a scalar some scalar comparison must be made. For spacecraft problems, angular momentum is the best quantity to use for this purpose. In general, any quantity that is conserved is a good choice for choosing the step size. General-purpose solvers do not know anything about the problem so they use the entire state for error correction.

Some solvers can also vary the order of the method. With Runge–Kutta this can be done every step since a step is entirely self-contained, i.e., step k does not rely on step $k - 1$.

A.2.3 Discontinuities

Many mechanical systems have nonlinearities that are modeled using discontinuous functions. Coulomb friction is an example. Coulomb friction is shown in Example A.2.

```
1 Plot2D([-1 0 0 1],[-1 -1 1 1],'Velocity','  
    Friction','Coulomb_Friction')  
2 set(gca,'ylim',[-1.5 1.5]);
```



Example A.2: Coulomb friction.

When numerically integrating a function with Coulomb friction, the solution will limit cycle, that is manifest a self-sustained oscillation. This would imply that a mechanism with Coulomb friction would oscillate forever. Rather than using this kind of function, the discontinuous function can be replaced with a smooth function. For example, the function

$$y = \frac{2}{1 + e^{-bx}} - 1.0 \quad (\text{A.47})$$

produces Example A.3. If the device never passes through zero then the discontinuous model is fine. If it does, you may need another numerical model.

A.3. Fourier series

Fourier series are a useful way to study the behavior of a system as a function of frequency. This section will describe Fourier series using trigonometric functions.

A.3.1 Trigonometric identities

Trigonometric sum of angle formulas and sum of squares formulas.

$$\sin(a \pm b) = \sin a \cos b \mp \sin b \cos a \quad (\text{A.48})$$

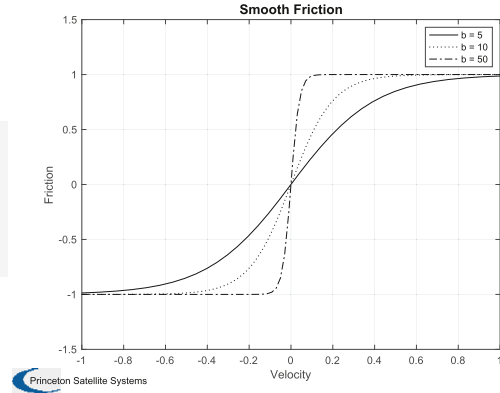
$$\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b \quad (\text{A.49})$$

$$1 = \sin^2 a + \cos^2 a \quad (\text{A.50})$$

```

1 x = linspace(-1,1);
2 y0 = 2./(1 + exp(-5*x)) - 1;
3 y1 = 2./(1 + exp(-10*x)) - 1;
4 y2 = 2./(1 + exp(-50*x)) - 1;
5 Plot2D(x,[y0;y1;y2], 'Velocity', 'Friction', 'Smooth Friction');
6 legend('b_0=5', 'b_0=10', 'b_0=50');
7 set(gca, 'ylim', [-1.5 1.5]);

```



Example A.3: Smooth friction.

You can get many useful relationships from these. For example:

$$\cos(2a) = \cos^2 a - \sin^2 a = \cos^2 a - (1 - \cos^2 a) = 2\cos^2 a - 1 \quad (\text{A.51})$$

$$\cos^2 a = \frac{1}{2} (1 + \cos(2a)) \quad (\text{A.52})$$

A.3.2 Sine and cosine Fourier series

If the function is periodic over $-\pi$ to π .

$$f(x) = a_0 + \sum_{k=1}^{\infty} (a_k \cos nx + b_k \sin nx) \quad (\text{A.53})$$

where

$$a_0 = \frac{2}{\pi} \int_{-\pi}^{\pi} f(x) dx \quad (\text{A.54})$$

$$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx \quad (\text{A.55})$$

$$b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx \quad (\text{A.56})$$

If the function is periodic over the interval $-L$ to L let $t = \frac{Lx}{\pi}$. The Fourier series is now

$$f(x) = a_0 + \sum_{k=1}^{\infty} \left(a_k \cos \left(n \frac{\pi t}{L} \right) + b_k \sin \left(n \frac{\pi t}{L} \right) \right) \quad (\text{A.57})$$

where

$$a_0 = \frac{2}{L} \int_{-L}^L f(x) dx \quad (\text{A.58})$$

$$a_k = \frac{1}{L} \int_{-L}^L f(x) \cos\left(n \frac{\pi x}{L}\right) dx \quad (\text{A.59})$$

$$b_k = \frac{1}{L} \int_{-L}^L f(x) \sin\left(n \frac{\pi x}{L}\right) dx \quad (\text{A.60})$$

A.4. Spherical geometry

Spherical geometry is an alternative way of looking at unit vectors. Fig. A.2 shows the unit sphere and three unit vectors. There are two angles that are important in spherical

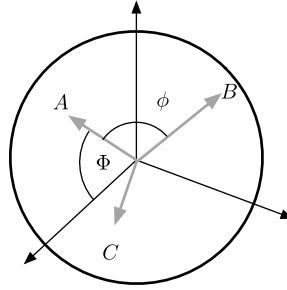


Figure A.2 Unit sphere and unit vectors.

geometry. The first is the angle ϕ between the vectors, which is found from the dot product,

$$A^T B = \cos(\phi) \quad (\text{A.61})$$

which is the dot product. The second angle is the angle Φ between the arcs, for example angle Φ between arcs AC and AB . Spherical trigonometry is useful when using vectors to stars or planets for measurements. Two of the most important equations are the law of sines and law of cosines for spherical angles. The law of sines is

$$\frac{\sin(\theta)}{\sin(\Theta)} = \frac{\sin(\lambda)}{\sin(\Lambda)} = \frac{\sin(\phi)}{\sin(\Phi)} \quad (\text{A.62})$$

The law of cosines for the sides is

$$\cos(\lambda) = \cos(\theta) \cos(\phi) + \sin(\theta) \sin(\phi) \cos(\Lambda) \quad (\text{A.63})$$

and the law of cosines for the angles is

$$\cos(\Lambda) = \cos(\Theta) \cos(\Phi) + \sin(\Theta) \sin(\Phi) \cos(\lambda) \quad (\text{A.64})$$

The small Greek letter denotes the angle from the dot product. The capital Greek letter denotes the angle between the arcs on the surface of the sphere that is opposite the corresponding small letter. The three arcs form a spherical triangle.

A.5. The chain rule in calculus

The chain rule often appears in derivations

$$\frac{dab}{dt} = a \frac{db}{dt} + \frac{da}{dt} b \quad (\text{A.65})$$

Sometimes we want to change the independent variable in a differential equation. For example,

$$\frac{dx}{dt} + \sigma x = u \quad (\text{A.66})$$

Suppose we want to differentiate with respect to θ and $\theta = \omega t$. If ω is a constant then

$$d\theta = \omega dt \quad (\text{A.67})$$

or

$$\frac{dx}{d\theta} + \sigma \frac{x}{\omega} = \frac{u}{\omega} \quad (\text{A.68})$$